

(“,”) for the same.

- *fill*: Takes a Boolean value to indicate whether to insert blank values when there are rows of unequal length.
- *comment.char*: Represents the character to be used to ignore comment lines. The default value is hash (“#”).

You can get a count of the number of rows and columns in the data frame using the *nrow()* and *ncol()* functions respectively. For the bank data set, we have the following values:

```
> nrow(bank)
[1] 4521
> ncol(bank)
[1] 17
```

The *head()* and *tail()* functions on the data frame provide the first and last six records respectively. See Figure 1.

You can obtain the minimum and maximum values for the ‘age’ field using the following functions:

```
> min(bank$age)
[1] 19

> max(bank$age)
[1] 87
```

The *summary()* function provides information on class, mode, median, minimum, maximum for the various fields, as

illustrated in Figure 2.

```
> print(summary(bank))
   age          job          marital          education
Min.   :19.00   Length:4521   Length:4521   Length:4521
1st Qu.:33.00   Class :character Class :character Class :character
Median :39.00   Mode  :character Mode  :character Mode  :character
Mean   :41.17
3rd Qu.:49.00
Max.    :87.00

 default          balance          housing          loan
Length:4521      Min.   : -3313   Length:4521   Length:4521
Class :character 1st Qu.:   69   Class :character Class :character
Mode  :character Median : 444   Mode  :character Mode  :character
Mean   : 1423
3rd Qu.: 1480
Max.    :71188

 contact          day          month          duration
Length:4521      Min.   : 1.00   Length:4521   Min.   : 4
Class :character 1st Qu.: 9.00   Class :character 1st Qu.: 104
Mode  :character Median :16.00   Mode  :character Median : 185
Mean   :15.92
3rd Qu.:21.00
Max.    :31.00

 campaign          pdays          previous          poutcome
Min.   : 1.000   Min.   : -1.00   Min.   : 0.0000   Length:4521
1st Qu.: 1.000   1st Qu.: -1.00   1st Qu.: 0.0000   Class :character
Median : 2.000   Median : -1.00   Median : 0.0000   Mode  :character
Mean   : 2.794   Mean   : 39.77   Mean   : 0.5426
3rd Qu.: 3.000   3rd Qu.: -1.00   3rd Qu.: 0.0000
Max.    :50.000   Max.    :871.00   Max.    :25.0000

 y
Length:4521
Class :character
Mode  :character
```

Figure 2: Output of *summary()*

You can obtain a portion of the data that satisfies condition(s) using the *subset()* function. For example, you can find those who belong to *Management*, are married, have both housing and personal loans, and still maintain an annual balance of greater than 5000 Euros using the following construct:

```
> head(bank)
   age          job marital education default balance housing loan contact day
1  30 unemployed married primary         no   1787      no   no cellular 19
2  33   services married secondary        no   4789     yes   yes cellular 11
3  35 management single  tertiary        no   1350     yes   no cellular 16
4  30 management married tertiary        no   1476     yes   yes unknown  3
5  59 blue-collar married secondary        no     0     yes   no unknown  5
6  35 management single  tertiary        no    747      no   no cellular 23

 month duration campaign pdays previous poutcome y
1  oct         79         1      -1         0 unknown no
2  may        220         1    339         4 failure no
3  apr        185         1    330         1 failure no
4  jun        199         4     -1         0 unknown no
5  may        226         1     -1         0 unknown no
6  feb        141         2    176         3 failure no

> tail(bank)
   age          job marital education default balance housing loan contact
4516 32   services single secondary        no    473     yes   no cellular
4517 33   services married secondary        no   -333     yes   no cellular
4518 57 self-employed married tertiary      yes -3313     yes   yes unknown
4519 57  technician married secondary        no    295      no   no cellular
4520 28 blue-collar married secondary        no   1137      no   no cellular
4521 44 entrepreneur single tertiary        no   1136     yes   yes cellular

 day month duration campaign pdays previous poutcome y
4516  7   jul        624         5     -1         0 unknown no
4517 30   jul        329         5     -1         0 unknown no
4518  9   may        153         1     -1         0 unknown no
4519 19   aug        151        11     -1         0 unknown no
4520  6   feb        129         4     211         3   other no
4521  3   apr        345         2    249         7   other no
```

Figure 1: Output of *head()* and *tail()* function